

OPTICROP: SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE

SCALABILITY ROADMAP & FUTURE SCOPE

1. Planned System Enhancements

- IoT Integration: Connect soil sensors to collect NPK, pH, and humidity data automatically.
- Deep Learning Disease Detection: Build a PyTorch CNN to identify crop leaf anomalies from images.
- Weather API Integration: Integrate weather forecast services to schedule irrigation automatically.
- SaaS Multi-tenancy: Add authentication, subscriptions, and user management for agribusinesses.